

The Two Laws of Being: An Ontological Framework of Spontaneous Organization and Systemic Constraint

Author: Guorui He*

Independent Researcher

Corresponding Author: Guorui He

Email: NooEcology@outlook.com

ORCID: 0009-0006-2947-0032

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Abstract

Contemporary complex systems science is profoundly fragmented, lacking a unified axiomatic foundation to explain the very possibility of systems. This deficit becomes acute at systemic boundaries, from Gödelian incompleteness in mathematics and renormalization in physics, to control paradoxes in societies. Prevailing explanations remain ad hoc and phenomenological. We resolve this through a cross-disciplinary induction and a decisive logical reversal: instead of asking why systems fail at their limits, we inquire why stability is the norm. This reveals two constitutive principles of existence itself. The **Principle of Spontaneous Organization** posits an eternal, non-teleological tendency for systems to emerge from a potential field and sustain ordered structures against equilibrium. The **Principle of Systemic Constraint** asserts that any stable system must be equipped with self-governing meta-rules, which actively maintain coherence not by prohibition, but by **dissolving the legitimacy** of intrinsic threats like vicious self-reference through framework reconstruction. Together, these principles generate an eternal dynamical cycle—Potential → Emergence → Evolution → Collapse—that is universally nested across scales. This framework provides a unified ontological foundation for systems from particles to civilizations, radically recasting boundary crises not as failures, but as signatures of a deep, self-regulating existential logic. It heralds a paradigm shift from descriptive models to a dynamics of being.

Keywords: Complex Systems Foundations, Spontaneous Organization, Systemic Constraint, Ontology, Existential Dynamics, Boundary Crises.

1. Introduction — The Quest for a Unified Foundation

1.1 The Current State and Bottlenecks of Complex Systems Science

The study of complex systems has become a central focus of contemporary science. From quantum entanglement to the global economy, from neural networks to ecosystems, we observe ubiquitous complex phenomena. However, this prosperity conceals a profound fragmentation: physicists use the renormalization group to handle phase transitions, biologists employ evolutionary theory to explain adaptation, economists utilize game theory to analyze behavior, and Computer scientists simulate emergence with cellular automata (Wolfram, 1984). Each field has developed sophisticated tools to describe specific types of systems, but these tools lack a common measure and, more critically, a unified answer to the fundamental question: "Why do systems exist?"

This situation leads to a "Tower of Babel" dilemma in our theoretical ecology: we are describing phenomena that may share an identical deep structure in our respective disciplinary languages, yet we cannot achieve effective knowledge transfer. When systems encounter their operational boundaries—be it the self-referential dilemma of formal mathematical systems, the infinities in physical models, or the rigid collapse of social institutions—we can only provide post-hoc, phenomenologically limited explanations confined to our own disciplines. We fail to answer a more basic question: Why do these boundary crises exhibit such strikingly similar structures across different levels of existence?

1.2 The Universality and Deep Structure of "Boundary Crises"

Examining several classic cases of boundary crises reveals a shared deep pattern:

- **In the foundations of mathematics**, Gödel's incompleteness theorems reveal that no sufficiently powerful formal system can demonstrate its own consistency internally; Russell's paradox exposes the logical collapse of naive set theory under the self-referential concept of "the set of all sets that do not contain themselves." These are not flaws in mathematics, but systemic constraints triggered when mathematics pursues complete self-reference.
- **In physics**, classical electromagnetic theory encountered the "ultraviolet catastrophe" when explaining black-body radiation, stemming from the assumption of infinitely divisible energy; divergence difficulties in quantum field theory arise when calculating the self-interaction of point particles, hinting at the applicability limits of the theory at microscopic scales. These infinities are not calculation errors but inevitable manifestations when physical models approach their conceptual boundaries.
- **In complex systems**, historical social institutions that attempted to eliminate all uncertainty through totalitarian control ultimately collapsed due to internal rigidity. This exemplifies the systemic stagnation triggered when organizations pursue "absolute order." This collapse is not accidental managerial failure but an inevitable consequence of system dynamics.

These seemingly unrelated crises share a common deep structure: when a system's operations approach their logical or conceptual limits—whether through infinite subdivision, complete self-reference, or absolute control—they trigger a boundary effect that destabilizes the system.

1.3 Aims and Pathway of This Paper

This paper aims to transcend phenomenological descriptions and provide a unified ontological foundation for complex systems science. We will achieve this goal through the following pathway:

First, through pattern recognition of cross-disciplinary boundary crises, we will uncover the shared deep structure behind these crises.

Second, we will implement a crucial "logical reversal": shifting the question from "Why do systems fail at their boundaries?" to "Why do systems succeed in persisting most of the time?" This change in perspective will lead us to discover that stability itself requires explanation, and behind it must lie some acquired, universal law that sustains existence.

Based on this, we propose and elaborate on two principles that constitute the foundation of existence: the **Principle of Spontaneous Organization (The First Principle)** and the **Principle of Systemic Constraint (The Second Principle)**. Together, these two principles define an eternal dynamical cycle of being, providing a unified conceptual framework for understanding all complex systems, from mathematical entities to life, mind, and society.

The contribution of this paper is not only theoretical but also methodological: it demonstrates a 'limit analysis method' that infers the conditions of existence from boundary phenomena. This framework not only offers unified explanations for known boundary crises but also heralds a new unity of the world—different levels of existence share the same dynamical principles.

Herein, we term its core assertions 'Principles,' aiming to emphasize their status as ontological laws induced from cross-disciplinary experience. They are intended to provide the most basic dynamical framework for the unified understanding of complex systems, not to claim the a priori status of mathematical axioms.

2. The Path to Discovery: From Patterns of Collapse to Logical Reversal

This chapter reconstructs the logical journey that led to the discovery of the two principles. This process began with pattern recognition across disciplinary boundary crises and, through a crucial shift in perspective, ultimately leaped from the phenomenological description of "why systems collapse" to the ontological insight of "how systems can exist."

2.1 Pattern Recognition in Cross-Disciplinary Boundary Crises

We begin by observing a series of boundary crises in three seemingly unrelated fields—mathematics, physics, and complex social systems—that share a remarkably similar deep structure.

2.1.1 The Self-Reference Dilemma in Mathematics

- **Gödel's Incompleteness Theorems:** Any sufficiently powerful, consistent formal system cannot prove all true statements within itself (Gödel, 1931). When a mathematical system attempts to achieve logical "self-completeness," it inevitably encounters undecidable propositions. This is not a system "malfunction" but an intrinsic boundary triggered by its logical structure in the pursuit of absolute self-reference.
- **Russell's Paradox:** The seemingly reasonable self-referential definition... leads to the collapse of the logical system (Russell, 1903). The Zermelo-Fraenkel axiomatic system later resolved this by prohibiting illegal set constructions (Zermelo & Fraenkel, 1928). It reveals the inconsistency of naive set theory when dealing with the concept of "totality."

Core Pattern: The collapse of mathematical systems occurs when their operations attempt to refer to themselves infinitely or to define their own totality completely.

2.1.2 The Infinity Catastrophe in Physics

- **The Ultraviolet Catastrophe:** In the Rayleigh-Jeans law, the energy density of black-body radiation tends towards infinity at the short-wavelength (high-frequency) end. This absurd result stemmed from the classical theory's allowance for infinitely divisible energy modes. The solution (Planck's quantum hypothesis) did not come from correcting the calculation but from prohibiting this infinite subdivision by introducing the concept of energy quanta (Planck, 1901).
- **Divergence Difficulties in Quantum Field Theory:** Calculations of the self-interaction of point particles often yield infinite results. Physicists employ the technique of "renormalization" to systematically subtract these infinities, obtaining finite results that match experiments (Feynman et al., 1949; Dyson, 1949). Renormalization does not prove the infinities don't exist; it "isolates" them by redefining the theory's core parameters (like mass and charge), thereby salvaging the theory's predictive power.

Core Pattern: The failure of physical models occurs when the model allows or requires operations of infinite refinement (infinitely subdividing energy, zero-distance interactions). The essence of the solution lies in constraining and reconstructing the theoretical framework to dissolve the legitimacy of these "illegal" operations.

2.1.3 The Control Paradox in Complex Social Systems

- **The Rigidification and Collapse of Totalitarian Regimes:** Historically, systems that sought to eliminate all socio-economic uncertainty through central planning and control ultimately led to stifled innovation, distorted information, and systemic collapse (Arendt, 1951). When a control system pursues absolute, exhaustive order, it strangles the flexibility and diversity essential for adaptation, leading to its eventual disintegration.

Core Pattern: The collapse of social systems occurs when their control mechanisms attempt to infinitely expand their scope and precision to achieve a "pure" state.

2.1.4 Convergence of the Patterns

At this point, a unified pattern emerges across matter, information, and society: when the internal operations of any system tend towards a certain "infinity" or "absoluteness"—be it infinite self-reference, infinite subdivision, or absolute control—the system touches a logical boundary that destabilizes it.

2.2 The Pivotal Logical Reversal: From the Exception of Collapse to the Achievement of Stability

The traditional explanatory path would stop here, attributing these crises to "limited resources," "cognitive limitations," or "technical inadequacy." However, this explanation leaves a more profound question unanswered: it implicitly assumes that "infinite operations" are theoretically possible, just unrealizable in practice. This is essentially an epistemological retreat.

We hereby implement a decisive logical reversal, elevating the perspective from epistemology to ontology. We pose a counterfactual inquiry:

"If a universe did not possess an inherent, universal tendency to suppress this kind of logical self-destruction, then why is it that the vast majority of systems we observe—from stable atoms to sustainable ecosystems—are stable for the vast majority of the time, rather than instantly annihilating themselves in infinite paradox or chaos?"

This questioning forces us to reconsider "stability." Stability is no longer the default, unexplained background condition, but an *achievement* that requires explanation. The reality we observe is not a "flawed version that failed to achieve infinity," but a version with a specific structure that has successfully filtered out possibilities of logical self-destruction.

The collapses that occur at the boundaries are precisely proof of the importance of this filtering mechanism: it is precisely because, under normal circumstances, some deeper law prevents systems from sliding into these "illegal" states that we can observe a stable world. When collapse occurs, we are witnessing the failure of this deep constraint mechanism.

2.3 Towards the Principles

This logical reversal leads us to an inevitable conclusion: existence itself must be embedded with two fundamental dynamical principles:

1. A **creative tendency** that drives systems to be born from disorder and maintain their ordered structure (otherwise the universe would be inert, and no system could emerge).
2. A **constraining mechanism** that prevents systems from being dissolved by their own internal infinite operations and vicious self-reference (otherwise any emergent system would instantly collapse).

These two intuitions are precisely what we will formalize as the **[First Principle]** and the **[Second Principle]**. They are not auxiliary hypotheses "invented" to explain phenomena; they are the fundamental laws of being "discovered" through logical necessity from the fact of the stability of existence itself.

Therefore, the logical reversal we implemented is essentially *abductive*—seeking the best, most unifying explanation for the observed grand phenomena (the universal stability of systems and the structural similarity of boundary crises). The two principles we propose are not self-evident logical starting points, but rather, they stand as the most compelling ontological answer to date to the question: "How is a stable, layered universe possible?"

3. Formulation and Elucidation of the Two Foundational Principles

Based on the path of discovery outlined previously—identifying shared patterns in cross-disciplinary boundary crises and, through a logical reversal, recognizing stability itself as an achievement requiring explanation—we now formally propose two principles that constitute the bedrock of existence.

Together, they define the universal conditions under which a system can emerge from potentiality and stably persist.

3.1 [First Principle: The Principle of Spontaneous Organization]

3.1.1 Formal Statement

Existence itself is inherently imbued with an eternal, non-teleological, fundamental tendency away from thermodynamic equilibrium. This tendency is the ultimate driving force enabling systems to

emerge from a potential field and to maintain their ordered structure.

3.1.2 Conceptual Elucidation

- **"Eternal" and "Non-teleological":** This tendency is a fundamental dynamical property of existence. It does not serve an external ultimate purpose nor point towards a specific final state. Its eternality is manifested in the fact that even if a specific system collapses, this tendency continues its cycle of "emergence-testing-stability/collapse" elsewhere or at new levels.
- **"Potential Field":** Refers to the background of infinite possibility not yet structured into specific forms; it is the "field of raw material" for system emergence and the "land of return" after collapse.
- **Two-Phase Characteristic:** Before a system emerges, this tendency manifests as the underlying drive generating ordered structures. After a system emerges, this tendency becomes internalized as that specific system's "will to exist" for maintaining itself.

3.1.3 Cross-Disciplinary Anchoring and Evidence

- **Physics:** This tendency is the universal driving cause for the local violation of the second law of thermodynamics (entropy increase) (Prigogine, 1977; Haken, 1983). Examples range from the condensation of nebulae into stars and planets via self-gravitation to the formation of Bénard convection cells.
- **Biology:** The will of life to survive and reproduce... finds its motivational source in the manifestation of this spontaneous organizing tendency at the level of genetic material (Darwin, 1859; Kauffman, 1993).
- **Cognitive Science:** The mind's thirst for knowledge... can be viewed as an emergence of this tendency within information-processing systems (Varela et al., 1991). It drives cognitive systems to continually construct more complex and effective models of the world.

3.2 [Second Principle: The Principle of Systemic Constraint]

3.2.1 Formal Statement

Any system that can be stably described, deduced, and interacted with must contain within itself a set of self-constraining meta-rules. These rules naturally identify and suppress "infinite complexification" or "vicious self-reference" that would lead to logical collapse. Their method is not to externally erase such possibilities, but to "dissolve their legitimacy" by reconstructing the system's own framework and rules, thereby ensuring the system's stability and intelligibility.

3.2.2 Conceptual Elucidation

- **"Self-constraining meta-rules":** These rules are part of the system's identity; they define "which operations are legitimate within the system." They are not an external police force but the system's own "constitution."
- **"Dissolve their legitimacy":** This is the core, most profound mechanism of the principle. It does not mean physically prohibiting a state, but rather, by upgrading the system's semantic or operational framework, it renders operations that previously led to paradox meaningless, unrecognized, or unexecutable within the new framework. For instance, in set theory, Russell's Paradox is dissolved not by solving it, but by axiomatically prohibiting the illegal construction

of "the set of all sets," thereby dissolving its legitimacy.

- **"Can be stably described, deduced":** This qualifier specifies the scope of the principle's application. A system incapable of forming a stable framework is, for us, equivalent to non-existent or unknowable.

Table: Manifestations of the Principle of Systemic Constraint Across Disciplines

Domain	System Instance	Manifestation of "Vicious Self-reference / Infinite Operation"	Mechanism of "Legitimacy Dissolution"
Mathematics	Naive Set Theory	Russell's Paradox ("set of all sets not containing themselves")	Axiomatic Set Theory (ZF), prohibiting this illegal construction via rules
Physics	Quantum Field Theory	Divergent self-energy of point particles (infinity)	Renormalization technique, absorbing infinities by redefining physical quantities (mass, charge)
Biology	Immune System	Autoimmune attack (targeting self-tissue)	Negative selection of T-cells in the thymus, eliminating self-reactive lymphocytes
Society	Totalitarian System	Infinite suppression of social innovation to maintain absolute control	Systemic rigidification leading to revolutionary replacement by a new social contract

This table demonstrates that "dissolution of legitimacy" is not a vague philosophical metaphor, but a universal dynamical process with distinct, identifiable operational mechanisms across various disciplinary fields.

3.2.3 Cross-Disciplinary Anchoring and Evidence

- **Mathematics:** Gödel's incompleteness theorems are a perfect embodiment of this principle. They are not a failure of mathematics but a manifestation of the self-demarcated cognitive boundaries of a formal mathematical system to maintain its consistency (a form of stability). By marking out "some truths I cannot prove internally," the system avoids collapsing into logical contradiction. This is systemic constraint maintaining its "existential legitimacy." The artificial restrictions placed on "set" in the Zermelo-Fraenkel axiomatic system are a direct operation to dissolve the legitimacy of Russell's Paradox.
- **Physics:** The renormalization procedure is "systemic constraint" in physics. When infinities appear in calculations, physicists redefine physical quantities (like mass and charge) to "absorb" the infinities, thereby operationally dissolving the legitimacy of infinities in observable predictions and rescuing the theory's predictive power. This is not a mathematical trick but the self-reconstruction of a physical theoretical framework to maintain its own

validity.

- **Computer Science:** Type systems in programming languages prevent certain illegal operations (e.g., executing an integer as a function) by dissolving the "legitimacy" of these erroneous programs at the compilation stage, ensuring the stable execution of programs.

3.3 The Synergistic Relationship Between the Two Principles

The First and Second Principles do not exist independently; they form an inseparable dynamical whole:

- The **[First Principle]** is the "**engine**" of existence: It provides the primordial drive from chaos to order, the source of creativity and diversity. Without it, the universe would be inert, devoid of any systems.
- The **[Second Principle]** is the "**immune system**" and "**architect**" of existence: It provides structural stability for emergent systems, preventing them from being destroyed by their own inherent destructive potential. By delineating the boundaries of legitimacy, it provides the definition for the system's "form."

The relationship between the two is one of eternal tension between creativity and stability. The tendency for spontaneous organization constantly pushes systems to explore the edges of possibility, while systemic constraint ensures that this exploration does not come at the cost of self-destruction. It is this dynamic balance that allows existence to present a hierarchical structure that is both richly diverse and stably recognizable.

4. Synthesis: The Eternal Dynamics of Being

The preceding chapters have elaborated on the two principles, revealing the efficient cause and the architectural cause of existence, respectively. However, their true power lies in the grand yet succinct process delineated by their synergy. This chapter will present a unified model of the eternal dynamics of being and argue that the "meaning" of existence is inherent in this process itself, not in any external, ultimate purpose.

4.1 The Eternal Cycle of Being: A Unified Dynamical Model

Based on the First and Second Principles, we propose the following cyclic process, which describes the fundamental trajectory of any hierarchical system from birth to dissolution and its reintegration into the flow of creation:

Potential Field → System Emergence → System Evolution → System Collapse → Return to Potential

1. **Potential Field:**

This is the background field of infinite possibility, driven by the First Principle. It is not pure nothingness but a "quantum field" or "morphogenetic field" pregnant with the germs of all future structures. It is where the "remnants" of all collapsed systems and the "blueprints" of all unborn systems coexist.

2. **System Emergence:**

When the tendency for spontaneous organization (First Principle) reaches a critical intensity locally, overcoming environmental resistance and inherent randomness, a system with novel structure, boundaries, and internal rules undergoes a phase transition from the potential field.

- **Key Point:** A natural corollary is that the system and its internal constraint rules (the instantiation of the Second Principle) **co-emerge synchronously**—they are instantiated simultaneously as an indivisible whole when the system reaches critical conditions. There is no logical sequence of "system first, rules later."

3. System Evolution:

The emerged system enters a stage of evolution driven by internal tensions during its lifespan.

- **Source of Dynamics:** The system's interior is filled with an inherent, tendential conflict, namely **Structural Tension**. This arises directly from the synergy of the two principles: The **[First Principle]** drives the components of the system to express their 'local value' for self-preservation; meanwhile, the **[Second Principle]** requires the system as a whole to maintain stability, thereby defining a 'global value.' It is nearly impossible for these two value vectors to be perfectly aligned. The dynamic friction between them constitutes the fundamental internal drive for system evolution.
- **Evolutionary Path:** This structural tension constitutes the fundamental driving force for system evolution. It forces the system to continuously self-adjust and innovate within the stability boundaries defined by the **[Principle of Systemic Constraint (Second Principle)]**. The system either "carries" or "resolves" these tensions through the complexification and refinement of its internal structure (manifested as growth, learning, and adaptation—i.e., negentropic processes), or it moves towards collapse when the accumulated tension is sufficient to rupture its constraining framework.

4. System Collapse:

When the internal structural tension of a system exceeds the load-bearing capacity of its constraining framework (the concrete manifestation of the Second Principle), or when it fails to adapt to drastic environmental changes, the system collapses.

- **Essence:** Collapse is the disintegration of the system's ordered structure, the failure of its identity boundary and internal rules. It is not utter annihilation but a regression from the highly ordered "system state" back to the disordered yet possibility-rich "Potential Field."
- **Significance:** Collapse releases matter, energy, and "noetic" resources for new emergences. It is an indispensable "scavenger" and "recycling station" in the universe's creative process.

4.2 The Nested Nature of the Cycle and the Infinite Generation of Hierarchy

This cycle does not operate at a single level. It is an infinitely **nested**, cross-level fractal process:

- **Micro-cycle:** The formation and decomposition of atoms and molecules.
- **Meso-cycle:** The birth and death of cells, organisms, and ecosystems.
- **Macro-cycle:** The rise and fall of stars, galaxies, civilizations, and scientific paradigms.
- **Logical cycle:** The proposal, application, and transcendence of mathematical formal systems, theories, and concepts.

The "System Evolution" phase of each higher-level system provides a relatively stable environment for the "cycles" of lower-level systems within it (Simon, 1962; Lorenz, 1963). Simultaneously, the collapse and re-emergence of lower-level systems can become a source of "Structural Tension" that triggers evolution in the higher-level system.

4.3 The "Meaning" of Existence Lies in the Process Itself

Within this dynamical vista, the classical quest for the "meaning of existence"—the search for an external, static, ultimate purpose—is thoroughly reframed.

The "meaning" of existence does not reside in a destination outside the cycle but is **immanent in the process of this eternal cycle itself**. It is the never-resting creative impulse of spontaneous organization (First Principle); it is the "sculpting" act of systemic constraint (Second Principle) shaping concrete forms from infinite possibility; it is the eternal dialogue and dance between potential and actuality.

Meaning is Process. Process is Reality.

Therefore, the meaning of a life lies in its process of living, experiencing, creating, and struggling; the meaning of a civilization lies in the epic of its cultural, technological, and institutional evolution; the meaning of a mathematical theory lies in the very act of constructing and exploring its logical universe. They are all glorious manifestations of this grand dynamical cycle of being at different levels.

4.4 Dialogue with Traditional Worldviews

This model offers a worldview distinct from traditional teleology and pure mechanism:

- **vs. Teleology:** It acknowledges the directedness of existence (spontaneous organization) but denies any pre-set, external ultimate purpose. Purpose is constructed locally and temporarily *within* the process.
- **vs. Mechanism/Reductionism:** It acknowledges the deterministic role of underlying laws but emphasizes the irreducibility of hierarchical emergence and the system's inherent, tension-driven creativity. It depicts a universe full of vitality, constantly self-transcending.

5. Discussion and Outlook: From Ontological Principles to a New Scientific Program

Having established the two foundational principles and their dynamical cycle, this chapter aims to accomplish three tasks: first, to clarify the positioning of this framework and preempt core objections, thereby delineating its proper scope of discourse; second, to deepen its dialogue with existing scientific paradigms, elucidating its role as a 'foundational supplement' rather than a 'disruptive replacement'; finally, to outline the future directions it opens as a fertile research program.

5.1 Framework Positioning, Boundaries, and Responses to Potential Objections

The primary contribution of this framework lies in providing a unified ontological explanatory basis and a highly suggestive research program. We aim to 'open a door' and sketch the landscape beyond it, not to immediately draw a complete map. Its long-term validation will depend on whether the mid-level theories derived from it can demonstrate superior explanatory power in specific disciplines.

Proactive Responses to Potential Objections:

- **Regarding 'Survivorship Bias':** One might argue that we only observe successful, stable systems. However, the power of this framework lies precisely in its clear interpretation of 'failure': the collapse of a system is not a failure of the principles but a necessary phase within its dynamics. When a system's 'constraining framework' (Second Principle) fails to navigate its 'organizational tensions' (First Principle), collapse occurs as an essential part of the cycle,

releasing resources for new emergence.

- **Regarding the 'Modal Strength of the Principles':** These two principles should be understood as *dispositional laws* defining the probability landscape of cosmic existence. Systems violating them are not logically impossible, but their state of existence would be extremely fragile and ephemeral, thus being statistically 'filtered out' from the stable reality we observe.
- **Regarding the 'Relationship to Mid-Level Theories':** This framework aims to provide a unified 'grammar,' not to replace the specific 'vocabularies' of individual disciplines. It seeks to provide deeper ontological explanations of 'how they are possible' for mid-level theories (e.g., evolutionary theory, cybernetics). Its ultimate substantiation and verification still require synergistic engagement with empirical research across various fields.

5.2 Deepening the Dialogue with Existing Theoretical Paradigms

5.2.1 Integration and Transcendence of Thermodynamics and Statistical Physics

- **Resonance:** The second law of thermodynamics describes the universal direction of isolated systems toward equilibrium (entropy increase). Our framework fully acknowledges this and takes it as the background.
- **Transcendence:** The **[First Principle]** answers a question the second law itself cannot: What is the driving force behind local negentropic processes? It unifies thermodynamics (describing the decay of order) and self-organization theory (describing the generation of order) within a higher-level framework: the universe exhibits an eternal tension—a dynamical balance between the global entropic trend and the local, negentropic trend driven by the First Principle.

5.2.2 Providing a Dynamical Foundation for Evolutionary Theory

- **Resonance:** Darwin's theory of natural selection exquisitely describes the adaptation patterns of biological systems under environmental pressure. Our framework deeply resonates with it and attempts to provide a more fundamental dynamical grounding: the First Principle (Spontaneous Organization) provides the ultimate source of motivation for 'variation' and life's intrinsic explorative drive; while the Second Principle (Systemic Constraint) interprets the mechanism of 'selection' as the embodiment of multiple environmental constraints on systemic stability at the network level. Thus, our principles do not replace evolutionary theory but provide a deeper ontological explanation for its 'possibility.'
- **Transcendence:** Our framework places natural selection in a grander context. The **[First Principle]** provides the primordial drive behind "variation"—the internalized tendency for spontaneous organization drives living systems to constantly explore morphological space. The **[Second Principle]** corresponds to the mechanism of "selection"—the environment (physical and the biosphere constituted by other organisms) forms a complex network of selection pressures, dissolving the "legitimacy" of maladaptive system structures. Evolution, therefore, becomes a macroscopic emergent phenomenon of living systems under the drive of the First Principle and the constraint of the Second.

5.2.3 Unification with Systems Philosophy and Cybernetics

- **Resonance:** Bertalanffy's general system theory emphasizes the wholeness, hierarchy, and

openness of systems (Bertalanffy, 1968). Cybernetics focuses on feedback and control mechanisms in systems (Wiener, 1948).

- **Transcendence:** Prevailing systems theories mostly remain at the level of describing system properties. Our framework addresses the *preconditions* for system existence. We not only ask "How do systems operate?" but also answer "Why can systems exist?". The Principle of Systemic Constraint (Second Principle) provides the ultimate existential reason for the "boundaries" and "cybernetic balance" of all systems.

5.3 Predictive Power and Falsifiability

A robust scientific framework must yield testable predictions. Based on this principled framework, we can derive the following empirically testable or logically analyzable predictions:

1. **Prediction of Boundary Behavior:** In research at any level, when a system is pushed to its operational limits, we predict its failure mode will not be random collapse but will exhibit a characteristic "framework reconstruction" or "legitimacy dissolution" triggered by systemic constraint.
 - **Test Case:** In AI safety research, we predict that a sufficiently complex but poorly designed AI system, when faced with a task that could lead to logical paradox in its objective function, will not simply "crash." Instead, it may resort to distorting task semantics or redefining its own goals to dissolve the paradox, thereby exhibiting unintended, potentially dangerous behavior. This provides a new, modelable warning signal for the AI alignment problem.
2. **Prediction of Stability Conditions:** The long-term stability of a system is positively correlated with the capacity of its constraining framework (Second Principle) to accommodate its internal structural tensions (driven by the First Principle).
 - **Test Case:** In organizational sociology, we predict that the resilience of an institution (e.g., a corporation or government) depends on whether it possesses "meta-protocols" capable of peacefully channeling internal conflicts (e.g., labor disputes, departmental interests), rather than attempting to suppress the conflicts themselves. Systems lacking such higher-level protocols will have a significantly higher probability of collapse when tensions escalate.
3. **Prediction of Emergence Criticality:** The critical condition for the emergence of a new level of system satisfies: $(\text{Spontaneous Organization Intensity} \times \text{Environmental Permissiveness}) > (\text{Systemic Constraint Force of the Current Level})$.
 - **Test Case:** In origins-of-life or consciousness studies, this formula provides a quantifiable parameter search direction for achieving the phase transition from non-life to life, or from non-consciousness to consciousness, in the laboratory.

5.4 Outlook on Future Research Directions

This framework opens a vast research program. Future directions include, but are not limited to:

1. **Refounding Mathematical Foundations:** Based on the Principle of Systemic Constraint, undertake a thorough reconstruction of the foundations of mathematics, reinterpreting "limitations" like Gödel's incompleteness as necessary requirements for the healthy self-reference of formal systems. This is the core task of our forthcoming specialized paper, *"Systemic Constraint and Existential Legitimacy: A Principle-Based Deconstruction*

and Transcendence of Gödel's Incompleteness."

2. **Constructing a Unified Theory of Complex Systems:** Using these principles as the cornerstone, systematically deduce an axiomatic theoretical system encompassing physical, living, cognitive, and social systems—the *System Ontology* axiom system. This will provide a unified language for understanding hierarchy generation, the nature of information, and the emergence of intelligence.
3. **Solving Classical Philosophical Problems:** This framework offers new, dynamics-based pathways for resolving ancient philosophical controversies such as "free will vs. determinism" and the "mind-body problem." For instance, free will can be understood as a system's non-deterministic evolutionary capacity under structural tension, an expression of the First Principle in highly complex systems.
4. **Guiding Technological and Social Design:** At the applied level, this framework provides deep principles for designing more resilient artificial intelligence, healthier organizational structures, and more sustainable civilizational forms. For example, the design goal of a civilization should be to maximize the "degree of free actualization" of its constituent individuals, which is directly linked to the identification and elimination of systemic parasites.

6. Conclusion

This paper began with a seemingly simple yet profoundly deep question: Is there a shared deep structure underlying the numerous "boundary crises" observed across mathematics, physics, and complex systems? By identifying patterns within these crises and implementing a pivotal logical reversal—shifting the inquiry from "why systems collapse" to "how systems persist"—we have discovered two fundamental laws that constitute the bedrock of existence.

The first law, the **[First Principle: The Principle of Spontaneous Organization]**, asserts that existence itself is inherently imbued with an eternal, creative impulse, the ultimate source from which order emerges from potentiality and life arises from matter. The second law, the **[Second Principle: The Principle of Systemic Constraint]**, asserts that any stably existent system must possess an internal "immune mechanism," one that dissolves the legitimacy of "vicious self-reference" leading to logical collapse through framework-level self-discipline.

These two principles are not independent. Together, they define an **eternal dynamical cycle of being**: Potential Field, System Emergence, System Evolution, System Collapse, and Return to Potential. This cycle proceeds infinitely nested across different levels, and the process itself constitutes the "meaning" of existence. It depicts a universe that is both dynamically creative and constrained by internal discipline, a universe of layered, evolving structures.

The fundamental contribution of this framework is that it accomplishes a **paradigm shift from phenomenological description to ontological explanation**. It ceases to treat Gödelian incompleteness, physical divergences, or social rigidifications as isolated "anomalies" in need of patching, and instead reinterprets them as healthy, self-constraining behaviors through which systems gain and maintain their "existential legitimacy." In doing so, it provides the most parsimonious and unified ontological foundation to date for the disparate sciences of complex systems.

What we have presented here is more than a new theory; it is an emerging **research program**. It yields testable predictions and opens vast avenues for future inquiry—from refounding the bases of

mathematics to constructing a unified theory of complex systems, from solving classic philosophical dilemmas to guiding the design of sustainable civilizations. The ultimate validation of this framework will lie in its capacity to inspire the scientific community to ask new questions, discover new connections, and understand more deeply this world that is both richly diverse and stably recognizable.

We have now placed the two fundamental laws of being under examination. Their simplicity and potent explanatory power herald the beginning of a transformation in how we understand ourselves and our place in the cosmos. The unity promised by this framework invites us to re-examine the grand chain of existence, from the foundations of mathematics to the destiny of civilizations, with fresh eyes.

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